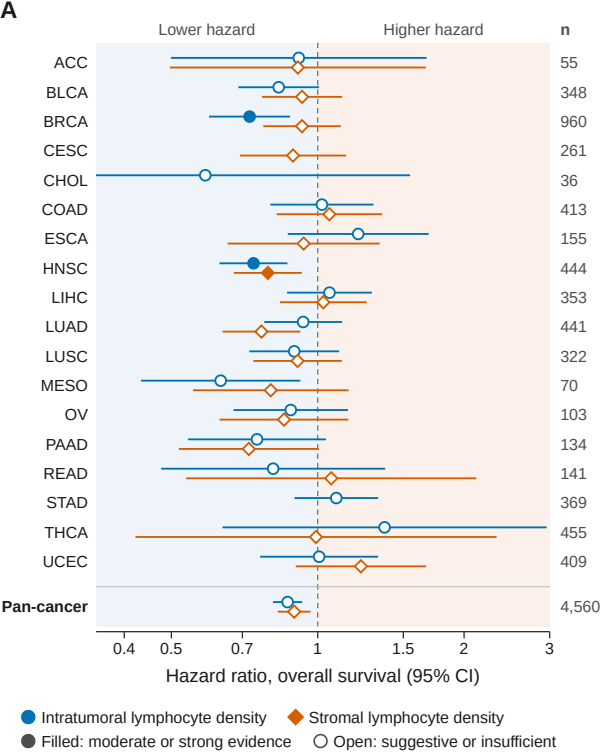
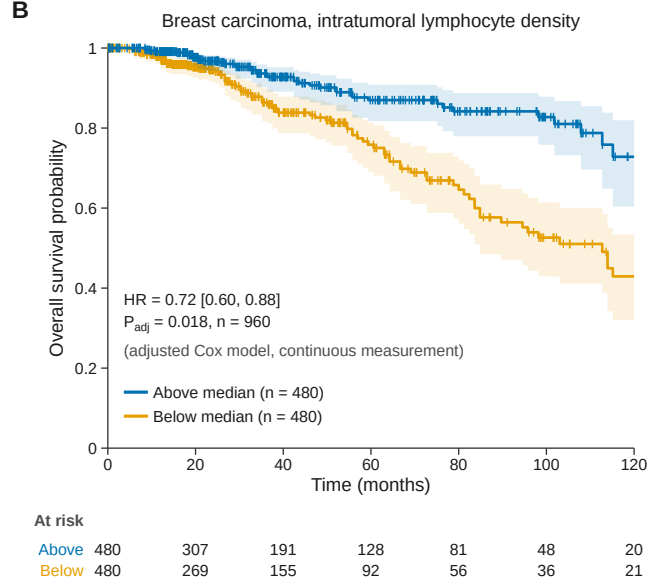
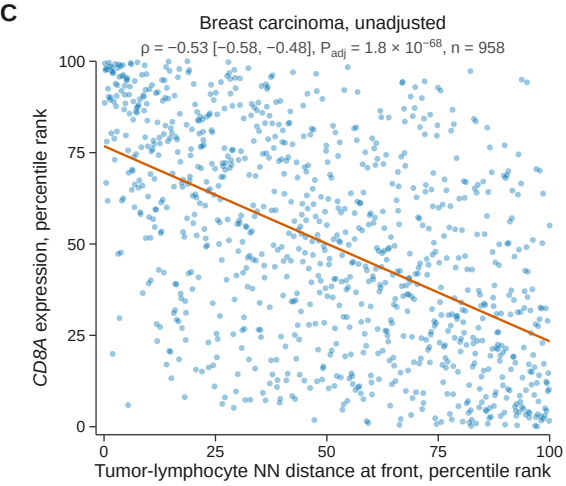
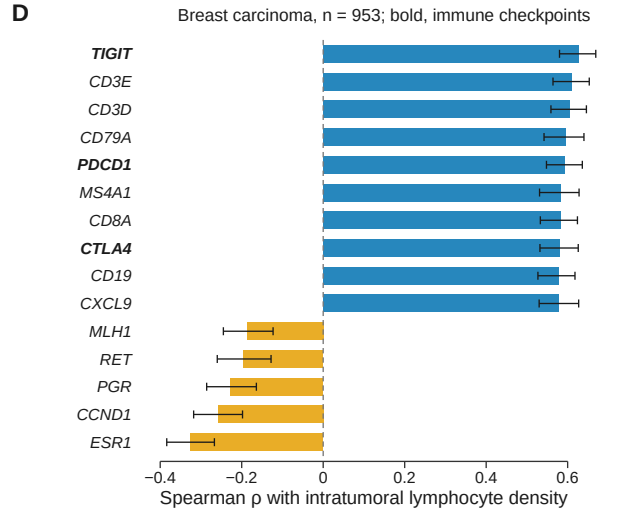

A
Lower hazard
Higher hazard
n
ACC 55
BLCA 348
BRCA 960
CESC 261
CHOL 36
COAD 413
ESCA 155
HNSC 444
LIHC 353
LUAD 441
LUSC 322
MESO 70
OV 103
PAAD 134
READ 141
STAD 369
THCA 455
UCEC 409
Pan-cancer 4,560
0.4 0.5 0.7 1 1.5 2 3
Hazard ratio, overall survival (95% CI)
Intratumoral lymphocyte density
Stromal lymphocyte density
Filled: moderate or strong evidence
Open: suggestive or insufficient


B
Breast carcinoma, intratumoral lymphocyte density
Overall survival probability
1
0.8
0.6
0.4
0.2
0
HR = 0.72 [0.60, 0.88]
P_adj = 0.018, n = 960
(adjusted Cox model, continuous measurement)
Above median (n = 480)
Below median (n = 480)
0 20 40 60 80 100 120
Time (months)
At risk
Above 480 307 191 128 81 48 20
Below 480 269 155 92 56 36 21


C
Breast carcinoma, unadjusted
ρ = −0.53 [−0.58, −0.48], P_adj = 1.8 × 10^−68, n = 958
CD8A expression, percentile rank
100
75
50
25
0
0 25 50 75 100
Tumor-lymphocyte NN distance at front, percentile rank


D
Breast carcinoma, n = 953; bold, immune checkpoints
TIGIT
CD3E
CD3D
CD79A
PDCD1
MS4A1
CD8A
CTLA4
CD19
CXCL9
MLH1
RET
PGR
CCND1
ESR1
−0.4 −0.2 0 0.2 0.4 0.6
Spearman ρ with intratumoral lymphocyte density